EXECUTIVE BUSINESS REPORT

Sales Forecasting & Demand Intelligence System

Prepared by: Vijay Tiwari | XYLOFY AI Internship | July 2026

Prepared for: Head of Supply Chain & CFO

1. EXECUTIVE SUMMARY

This report presents findings from an intelligent sales forecasting system built on 4 years of retail transaction data (2015–2018) covering 9,800 orders across 17 product categories in 4 regions of the United States.

The system answers three critical business questions every retail company faces: How much will we sell next month? Are there unusual demand patterns we should act on? And which products need different stocking strategies?

Using a combination of machine learning forecasting, anomaly detection, and product clustering, the system delivers actionable answers to all three — without requiring any technical knowledge to use, through a live interactive dashboard accessible from any browser.

2. KEY FINDINGS FROM DATA ANALYSIS

REVENUE PERFORMANCE

- Total revenue over 4 years: $2,261,537
- Technology is the top-earning category ($827,456 — 36.6% of revenue), followed by Furniture ($728,659) and Office Supplies ($705,422)
- The West region is the strongest performer with 21.4% average annual growth — nearly double the South region's 10.6%
- Overall business is growing consistently — monthly sales nearly doubled from ~$14,000 in January 2015 to ~$117,000 in November 2018

SEASONAL PATTERNS

- Sales follow a highly predictable yearly cycle:
  - November and December are always the strongest months (seasonal boost of +$33,000 to +$34,000 above average)
  - January and February are always the weakest months (seasonal dip of −$18,000 to −$27,000 below average)
  - September shows a consistent mid-year spike (+$27,676) likely driven by back-to-school and corporate Q3 budgets
- This pattern repeated identically across all 4 years — making it highly reliable for planning purposes

SHIPPING PERFORMANCE

- Average shipping time is 3.96 days across all regions
- Central region is slowest at 4.07 days — a marginal but consistent gap worth monitoring as order volumes grow

3. 3-MONTH SALES FORECAST (October — December 2026)

The forecasting system was tested against known historical data before being used to project future sales. The best-performing model (XGBoost Machine Learning) achieved an average error of 19.5% — meaning on a $100,000 month, predictions are typically within $19,500 of actual sales.

OVERALL FORECAST

Month	Forecasted Sales	Confidence Range
October	$69,215	$56,000 — $83,000
November	$87,072	$71,000 — $104,000
December	$64,997	$53,000 — $78,000

In plain terms: the business should prepare for its strongest quarter, with November expected to be the peak month. Total Q4 forecast is approximately $221,000 — up from $195,000 in the same period last year.

CATEGORY-LEVEL FORECAST (next 3 months combined)

- Office Supplies: $81,785 — most reliable forecast (7.8% error)
- Furniture: $76,107 — strongest growth trajectory (+145% Oct→Dec)
- Technology: $71,672 — moderate reliability (25.5% error)

4. TOP ANOMALIES DETECTED

The system scanned all 209 weeks of sales history and identified 11 confirmed unusual spikes — weeks where sales were more than 2 standard deviations above the 4-year average of $10,821 per week.

TOP 3 ANOMALIES:

① March 16, 2015 — $37,704 (Z-Score: +3.63)
LIKELY CAUSE: A large bulk corporate order, possibly a single institutional client purchasing office furniture or technology equipment in one transaction. This is the single largest weekly spike in the entire 4-year dataset.

② November 26, 2018 — $35,999 (Z-Score: +3.40)
LIKELY CAUSE: Black Friday and Cyber Monday week — classic festive season surge. Consumers and businesses stocking up on technology and office supplies during promotional periods.

③ November 12, 2018 — $30,572 (Z-Score: +2.67)

LIKELY CAUSE: Pre-holiday corporate procurement — companies placing large orders before year-end budget deadlines close.

KEY PATTERN: All confirmed anomalies are upward spikes — the business has never experienced a confirmed demand collapse. This is a positive signal: risk is on the upside (stockouts) not the downside (dead stock).

5. PRODUCT DEMAND SEGMENTATION

All 17 product sub-categories were grouped into 4 demand segments based on their sales volume, growth rate, demand predictability, and average order size.

SEGMENT 1 — HIGH VOLUME, GROWING 🟠

Products: Accessories, Binders, Chairs, Phones, Storage, Tables

These are the core revenue drivers — highest sales ($239K avg) with strong 25% annual growth. These products must be prioritised in inventory planning with 6–8 weeks of safety stock maintained at all times, especially entering Q4.

SEGMENT 2 — HIGH GROWTH, LOW VOLUME 🟢

Products: Appliances, Art, Labels, Paper

Growing fast (+44% annually) but currently low volume. Low risk since demand is predictable. Gradually increase stock each quarter to match the growth curve before demand outpaces supply.

SEGMENT 3 — HIGH VALUE, VOLATILE 🔴

Products: Copiers, Machines

Very large individual orders ($1,931 avg) but extremely unpredictable. Do not hold large standing inventory — use just-in-time ordering and maintain strong supplier relationships for rapid restocking when a large corporate order arrives.

SEGMENT 4 — STAGNANT / DECLINING 🔵

Products: Bookcases, Envelopes, Fasteners, Furnishings, Supplies

Near-zero or negative growth. Maintain lean inventory only. Consider promotional pricing to clear slow-moving stock and review this range for potential discontinuation in the next planning cycle.

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6. BUSINESS RECOMMENDATIONS

RECOMMENDATION 1 — BUILD A NOVEMBER STOCKOUT PREVENTION PROTOCOL

Data backing: November is the peak anomaly month (3 confirmed spikes across 4 years) AND the highest forecasted month ($87,072). The system predicts a surge but the actual spike could exceed this by 30-40% based on historical anomaly patterns.

Action: By September 1st each year, increase stock of Segment 3 products (Accessories, Phones, Chairs) by 40% above normal levels. Assign a dedicated procurement contact for emergency restocking.

RECOMMENDATION 2 — SHIFT WEST REGION INVESTMENT

Data backing: West region grew at 21.4% annually — nearly double Central (13%) and South (10.6%). West region forecast for Q4 is $80,557 — the highest of any region.

Action: Allocate a larger share of Q4 marketing and distribution budget to the West region. Consider opening an additional fulfilment point in the West to reduce the 3.93-day average ship time as volume continues to grow.

RECOMMENDATION 3 — AUTOMATE ANOMALY ALERTS

Data backing: All 11 confirmed anomalies exceeded $25,628 per week (the +2 standard deviation threshold). When this threshold is crossed, the business currently has no automated response.
Action: Set a weekly sales monitoring alert at $26,000. When triggered, automatically notify the procurement team to initiate an emergency restock review within 24 hours — before stockouts occur.

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7. SYSTEM LIMITATION

This forecasting system is built on historical patterns — it assumes the future will behave similarly to the past. It cannot predict:
- External disruptions (supply chain crises, economic downturns)
- New competitor activity or sudden market shifts
- One-off events like a single very large corporate order

The model's average error rate of 19.5% means forecasts should be treated as informed estimates with a ±20% confidence range — not precise guarantees. Business decisions, especially large inventory commitments, should always combine model output with human judgement from sales teams who have ground-level market context the model cannot capture.

Prepared by: Vijay Tiwari
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Live Dashboard: https://sales-forecasting-xylofy-ghw3vunntegpttxyhozxt8.streamlit.app
GitHub: https://github.com/vtgit04/sales-forecasting-xylofy
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